



Open & Reproducible Research

Research Methods in AI
2025 – 2026



Overview of this lecture

- Open data
- Computational reproducibility
- Replication
- Reproducibility in CS/AI
- Publication process



Open data

- Historically: scientific output = manuscripts
- Technology now allows us to easily share data, materials, notes, code, ...
 - Valuable and essential information in their own right
 - To evaluate scientific contributions
 - Reusability by other researchers

Open data

- Often required by scientific journals:

- **Nature:** "A condition of publication in a Nature Research journal is that authors are required to make materials, data, code, and associated protocols promptly available to readers without undue qualifications."
- **Science:** "After publication, all reasonable requests for data and materials must be fulfilled."



Open data



- ...and many top-tier AI conferences
 - **NeurIPS**: If you ran experiments, did you include the code, data, and instructions needed to reproduce the main experimental results
- **AAAI**: Reproducibility checklist
 - All novel datasets introduced in this paper are included in a data appendix
 - All source code required for conducting and analyzing the experiments is included in a code appendix

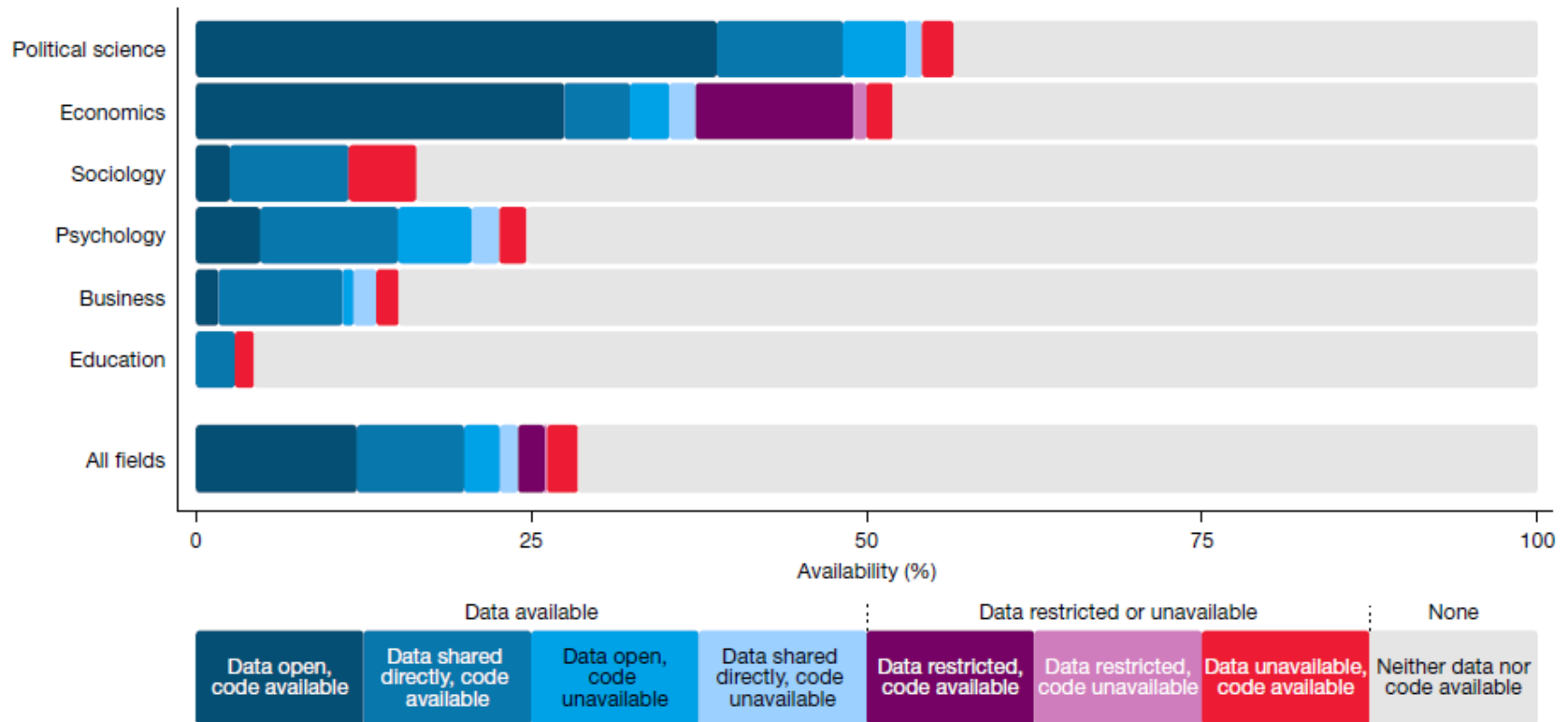


Open data

- Despite journal policies, many authors do not share data upon request
 - 27% according to Wicherts et al. (2006)
 - 38% according to Vanpaemel et al. (2015)



Open data



source: <https://www.nature.com/articles/s41586-026-10203-5>



Open data

- Reasons mentioned for not sharing:
 - No (longer) access to data
 - Lost
 - Computer crash
 - Collaborator left
 - Privacy and data sharing rules (see next week)
 - No time
 - Only when getting something in return
 - Not interested
 - ...



Open data

- Other considerations may apply as well:
 - Copyright & related (legal) issues
 - Some materials (e.g., certain questionnaires) can not be shared
 - Research within companies
 - Proprietary data or code



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Reproducibility

- Term used inconsistently across fields
- One definition: the possibility to obtain the same results as the originators of some specific scientific findings
- Umbrella term



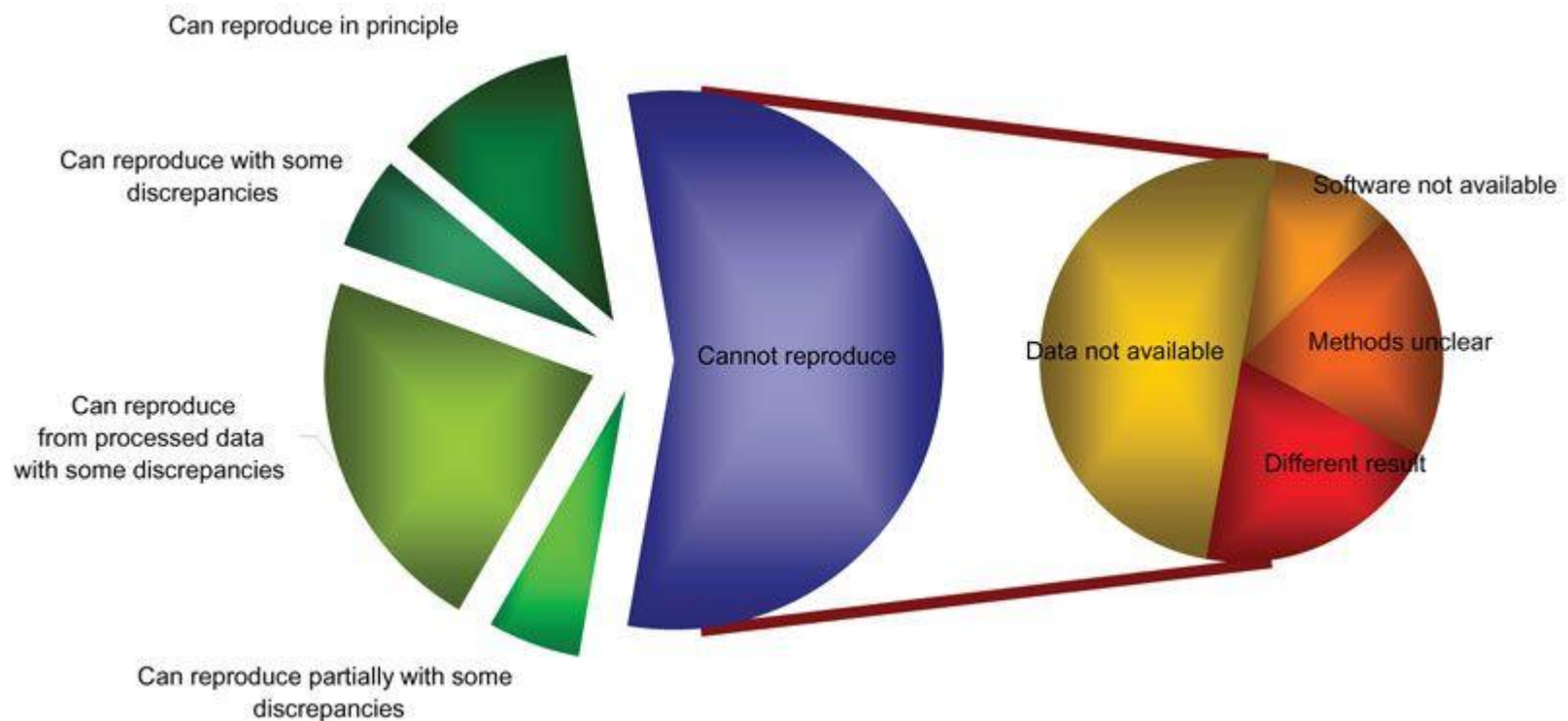


Computational reproducibility

- Definition: the possibility to obtain the same results and support for claims by using the same data and analytical procedures

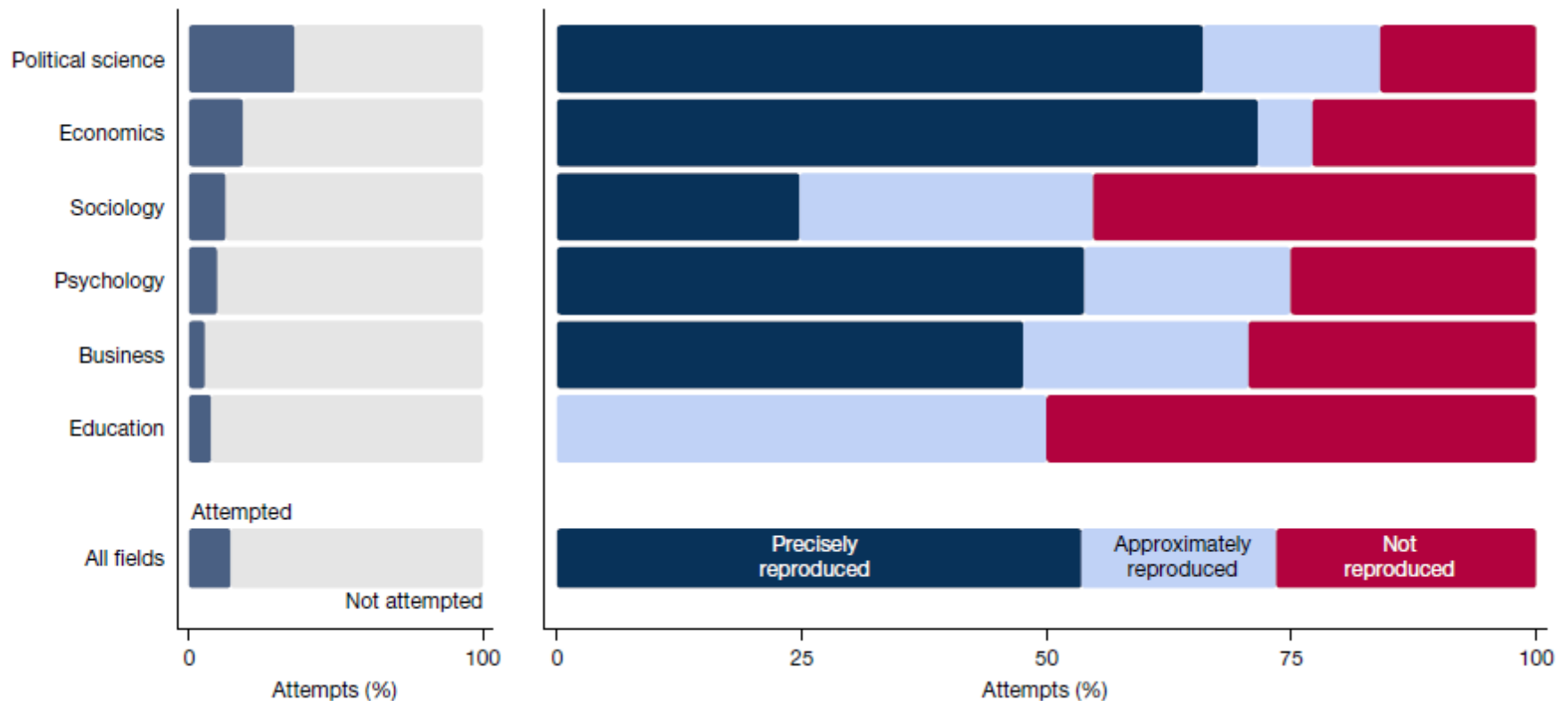
Computational reproducibility

■ Genetics (Ioannidis et al., 2009):



Computational reproducibility

- Social and behavioural sciences (Miske et al., 2026):





Exercise: spot the issues

We examined the effect of using AI-guided mindfulness & breathing coaching on people's stress level. To this end, we asked participants ($N = 10$) at the beginning of the study to rate on a seven-point likert scale how stressed they felt. In addition, we also measured their level of cortisol. Then, we instructed participants to use an app for AI-guided mindfulness & breathing coaching for month, and after this month, we again asked participants to rate their stress level on a seven-point likert scale and measured their cortisol. The results showed that people's subjective stress levels significantly reduced ($M_{\text{before}} = 5.40$, $M_{\text{after}} = 4.87$, $t(9) = 1.75$, $p < .05$). This finding supports the beneficial effect of using AI-guided mindfulness & breathing coaching on people's mental health.

Computational reproducibility

- Even if data are unavailable, you can assess reproducibility to some degree
 - Recalculated p-value from test statistic and degrees of freedom doesn't always match with the reported p-value (Bakker & Wicherts, 2011; Nuijten et al., 2016): <http://statcheck.io/>
 - Mean and SD of integer data is sometimes impossible given the sample size (Brown & Heathers, 2017): <https://steamtraen.shinyapps.io/rsprite/>



Computational reproducibility

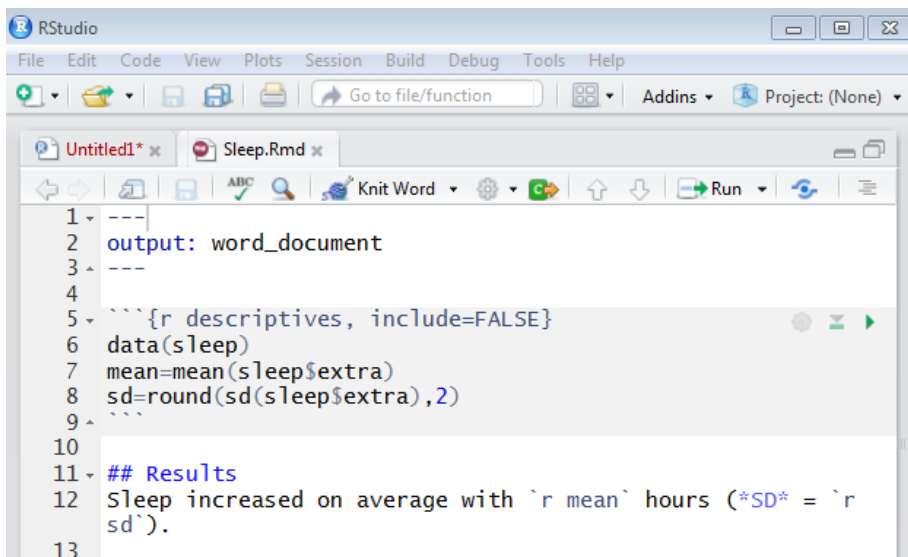
■ How do these errors arise?

- ☐ Copy-pasting
- ☐ Manual rounding
- ☐ Through several review-revision cycles
- ☐ Inadvertent omission of crucial information
- ☐ Questionable research practices
- ☐ Fraud
- ☐ No seed specified (e.g., for simulations)
- ☐ ...

Computational reproducibility

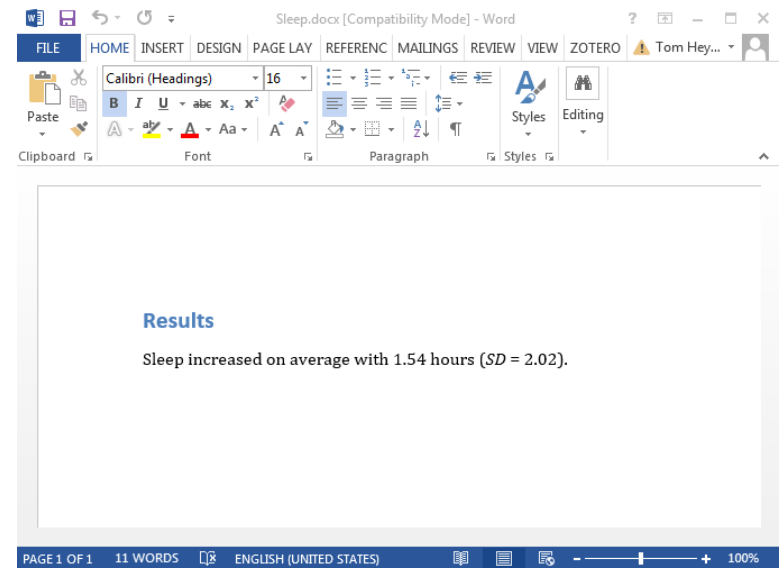
■ Some general solutions:

- Co-pilot (not the AI assistant, but could be used too)
- Reproducibility check by journals/reviewers (see later)
- Literate programming inspired tools: Jupyter Notebook, Rmarkdown, Quarto,...



The screenshot shows the RStudio interface with a script editor containing R code. The code is as follows:

```
1 ---
2 output: word_document
3 ---
4
5 {r descriptives, include=FALSE}
6 data(sleep)
7 mean=mean(sleep$extra)
8 sd=round(sd(sleep$extra),2)
9
10
11 ## Results
12 Sleep increased on average with `r mean` hours (*SD* = `r
13 sd`).
```





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Replication

- Definition: The ability of a researcher to duplicate the results of a prior study if the same procedures are followed but new data are collected (U.S. National Science Foundation)

Replication prevalence

- Targeted replication studies are rare
 - ~ 0.2 - 1.07% in Psychology (Clarke et al., 2026, Makel et al., 2012)
 - ~ 0.1% in Economics (Mueller-Langer et al., 2019)
 - ~ 0.02% in Ecology and Evolution (Kelly, 2019)
 - < 0.01% in Software Engineering (da Silva et al., 2012)

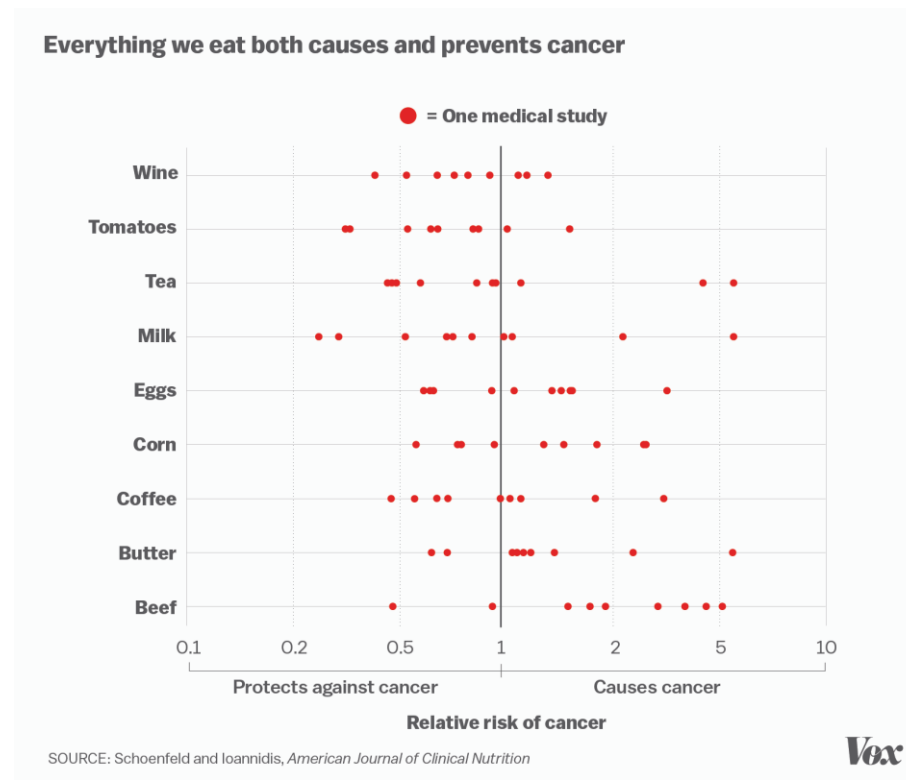


Replication prevalence

- Why so few replication studies?
 - Replication studies tend to have a bad reputation
 - If successful => nothing new
 - If unsuccessful => what went wrong?
 - Personal attacks
 - Can be costly to undertake
 - In some fields, replications (and computational reproducibility checks) are not feasible/possible
 - Not enough resources: computing power, time, money
 - Information not available

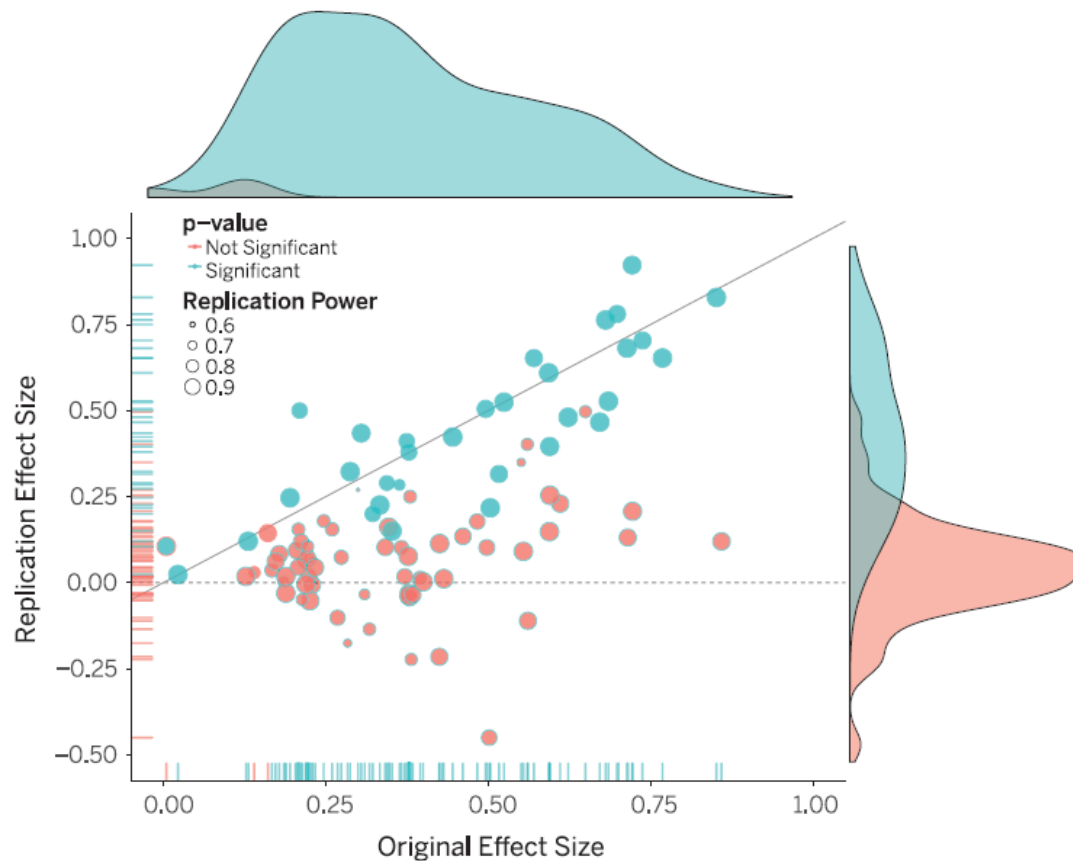
Replication success

- Nutritional epidemiology (Schoenfeld & Ioannidis, 2013):



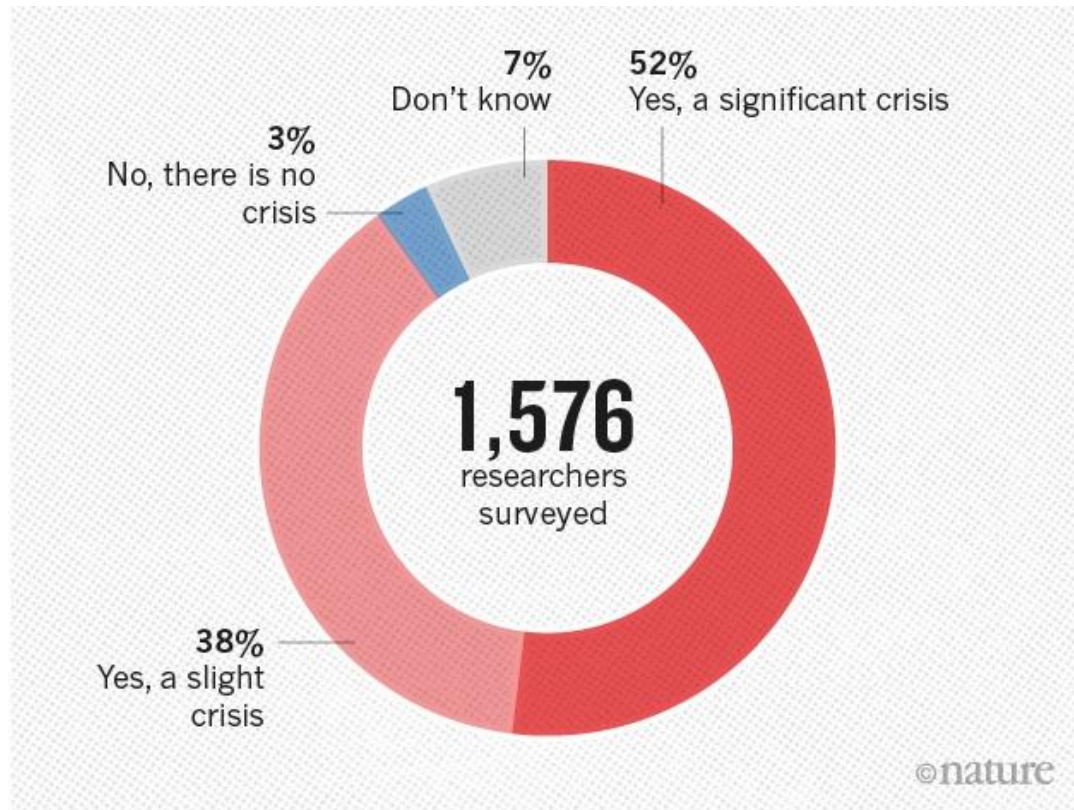
Replication success

- Psychology (Open Science Collaboration, 2015):



Replication success

■ Is there a replication crisis?



Data from 2016



Replication success

- General explanations for low success rate:
 - Fraud
 - Questionable research practices
 - Different procedures (due to incomplete reporting)
 - Poor computational reproducibility
 - ...



Replication

- Some solutions:

- ☐ Incentives (both for prevalence and success)
- ☐ Training
- ☐ Reporting standards
- ☐ Peer-review standards



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- **Reproducibility in CS/AI**
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Reproducibility in CS/AI

- Umbrella term reproducibility is used, because delineation between computational reproducibility and replication might not always be clear in computer science/AI
 - Library updates
 - Hardware differences
 - What is “new” data?
 - ...

Reproducibility in CS/AI

- AI (Gundersen & Kjensmo, 2018):
 - “Between a fifth and a third of the variables required for reproducibility are documented”
- Computer Science (Simkó, in prep):
 - One third of links to public repositories (e.g., Github) were broken or private
 - Model weights were shared in less than one-third of cases
 - Documentation quality varied widely, often limited to minimal readme files
 - Dependency declarations were inconsistent or missing in nearly half of cases



hxy-62 on Feb 3

Author ...

Hey, can you be more specific about what you try to do? I could help with clarifying things.

thank you for your quick reply. I just wanna use the concept500 dataset to test the sae feature steering performance. I have read this paper many times, but it really confuse me how to use this dataset to evaluate sae steering.

I don't need to train any thing, just eval the performance of sae feature for steering.



frankaging on Feb 3

Member ...

Got it. Assuming you have access to cursor or similar coding agent IDE:

Here are some prompts will get you to achieve your task:

- Since you have trained SAE vectors, you can skip generate.py and train.py. They are for generating the training datasets, and train supervises dictionary learning methods.



frankaging on Feb 4

Member ...

here is one 'metadata.jsonl': axbench/concept500/prod_9b_l20_v1/generate/metadata.jsonl

you can find others in the same folder. note that the main reason the system complains NOT FOUND is because we are bypassing training here since you are using pretrained SAEs and want to evaluate them directly. So you sort of want to create some "fake" folders in order to bypass those checks. does that make sense?

you can also run train.py with your SAEs. i think the code will just create that "fake" folder for you and generate metadata.jsonl along with some artifacts.

i am very surprised that coding agent could not figure this out nowadays. i will prompt my coding IDE to do some refactoring over the next couple of days if you can wait.

Reproducibility in CS/AI

- Often-used approach: data splitting
 - A model is trained to fit the training data, checked on the validation set, and evaluated on the test set



- Some parallels with replication
 - You do one thing on the training and validation, and then see whether it generalizes to the test set

Reproducibility issues in CS/AI

- Data splitting issues:

- Confounds

- e.g., “we took the last 10% as our test set”



- ⇒ Solutions:

- ⇒ Verify that properties of intended splits are (approximately) the same
 - ⇒ Random shuffling before splitting data

Reproducibility issues in CS/AI

- Data splitting issues:

- Underspecification

- e.g., “we divide the images into three parts: training, validation, and test subsets with the ratio of 50%, 25%, 25%
 - If the data is available as a whole (without the splits), not reproducible
 - No insight into why a certain percentage is chosen

⇒ Solution: Report train, validation and test set + code on how they were created, so transparency



Reproducibility issues in CS/AI

■ Data splitting issues:

- Leakage: information from validation/test set “leaking” into training
 - e.g., preprocessing before splitting
 - e.g., overlapping data

⇒ Solutions:

- ⇒ Split data before doing anything else
- ⇒ Careful inspection of data

Reproducibility issues in CS/AI

■ Metric-related issues:

□ Class imbalance

	Spam	Not spam
Marked as Spam	30 (True positive)	10 (False positive)
Marked as not spam	20 (False negative)	140 (True negative)

- $\text{Accuracy} = (30 + 140) / (30 + 140 + 10 + 20) = .85$
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 30 / (30 + 10) = .75$
- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 30 / (30 + 20) = .60$
- $\text{F1} = 2 * \text{Prec} * \text{Rec} / (\text{Prec} + \text{Rec}) = .67$

Reproducibility issues in CS/AI

■ Metric-related issues:

□ Class imbalance

	Spam	Not spam
Marked as Spam	30 (True positive)	10 (False positive)
Marked as not spam	20 (False negative)	40 (True negative)

- Accuracy = $(30 + \mathbf{40}) / (30 + \mathbf{40} + 10 + 20) = \mathbf{.70}$
- Precision = $TP / (TP + FP) = 30 / (30 + 10) = .75$
- Recall = $TP / (TP + FN) = 30 / (30 + 20) = .60$
- F1 = $2 * Prec * Rec / (Prec + Rec) = .67$



Reproducibility issues in CS/AI

- Metric-related issues:

- Class imbalance

- Matters a lot depending on the metric

- ⇒ Solutions:

- ⇒ Mindful when using accuracy
 - ⇒ Use other metrics instead/as well



Reproducibility issues in CS/AI

- Issues regarding non-deterministic processes:

- Seed selection

- Different outcome depending on the particular seed

- ⇒ Solution:

- ⇒ Seed averaging: use different seed and average performance



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- Reproducibility in CS/AI
- **Publication process**

Step 1: Identify suitable outlet

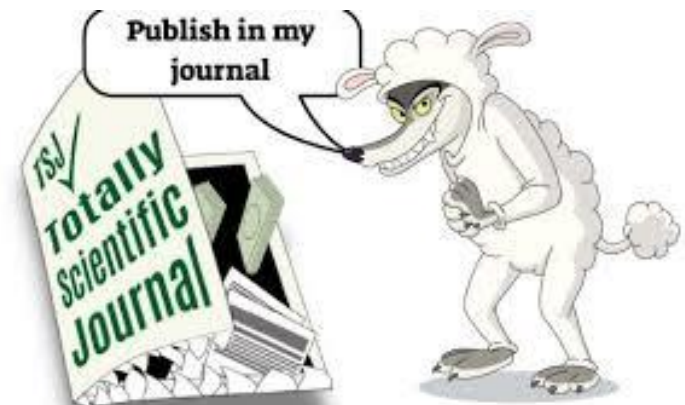
■ Be wary of

☐ Predatory journals

- No legit quality control
- Fast publication times
- Charging a (small) fee

☐ Paper mills

- Organizations producing “custom” academic papers
- Typically, plagiarism or fabrication



Step 1: Identify suitable outlet

■ From my spam folder:

Dear Doctor,

Greetings, we invite you to submit new articles to the **International Journal of Nursing and Health Care Research**. We are offering a *[Special issue fee of USD 3000 for five articles. The fee for one article is USD 1060 \(today's offer\)](#)*. If your article is ready, kindly submit it via this email or through our online [submission form](#).

We look forward to your response.

Regards,

Julia Quinn | juliaquinn.jour21@gmail.com

Editorial Coordinator

ISSN: 2688-9501, Volume 09, Impact Factor:4.64

International Journal of Nursing and Health Care Research.

Dear Author,

ARCHEO SCIENCES invites concise, high-quality manuscripts from all disciplines under its Multidisciplinary scope.

Indexed in: Web of Science

Broad Scope: Accepting papers from all subject areas

Submit your paper to: editor@archeosciences.org

Dear Professor,

We are thrilled to invite you to share your research in our Medical and Pharma journals. Your work is vital to advancing medical knowledge and improving patient care.

Our journals focus on publishing rigorous research that informs clinical practices and pharmaceutical advancements. By contributing your insights, you will connect with a global audience of healthcare professionals and researchers.

You can select the journal related to your research interest in this below link: <https://www.jmedarchives.com/>

I appreciate your submission before **April 10th, 2026**. Kindly acknowledge your submission with a mail.

Get me off Your Fucking Mailing List

David Mazières and Eddie Kohler

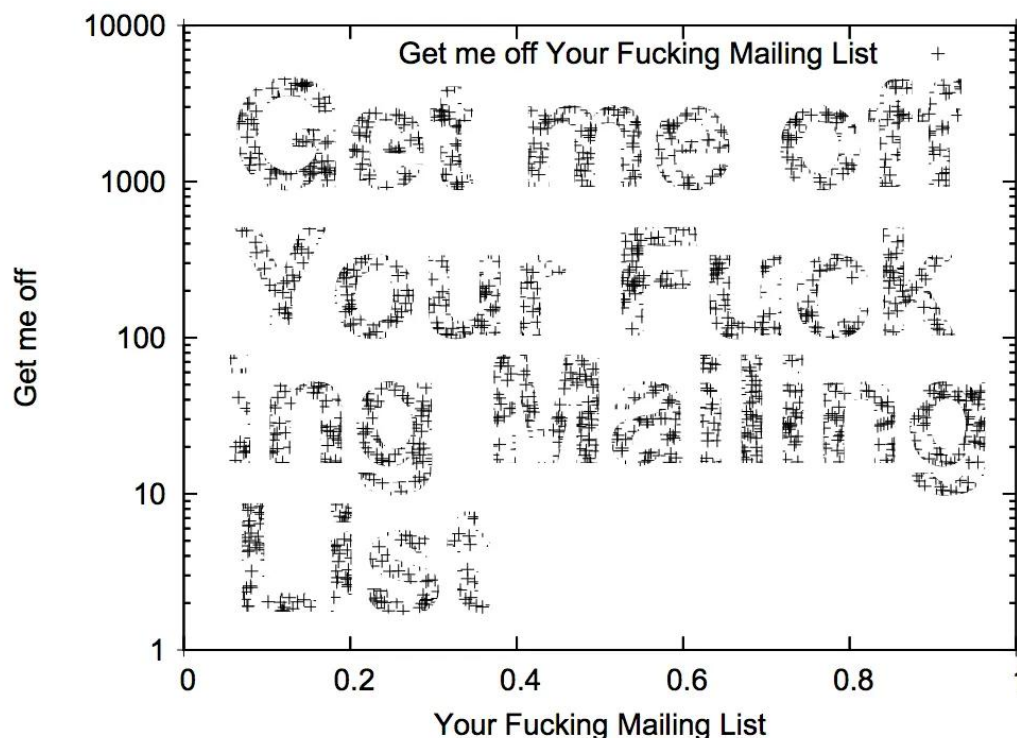
New York University

University of California, Los Angeles

<http://www.mailavenger.org/>

Abstract

Get me off your fucking mailing list.



Get me off your fuck-off your fucking mailing list. Get me off your fucking mailing list. Get me off your fucking mailing list.

Figure 2: Get me off your fucking mailing list.

Step 2: Editor assesses fit

- Can lead to so-called desk reject => researchers need to look for different venue/outlet (after revising the paper) or not publish the paper
- Or paper is sent to one to five peer-reviewers
 - Can be difficult to find experts willing to do this
 - Might be a reason to reject a paper

1. "Despite many invitations sent to possible reviewers, I (the Editor) was not able to collect a reviewer report so far. I am very sorry about that. In order not to block further possible publication of your study elsewhere and not start a never-ending story, I have taken the decision to reject the paper".



Step 3: Peer-review

■ Different types

- ☐ Single-blind: reviewer(s) remain anonymous
- ☐ Double-blind: both reviewer(s) and author(s) remain anonymous
- ☐ No blinding: identities of both reviewer(s) and author(s) are known to each other



Step 3: Peer-review

- Reviews may remain private or can be published openly
- Reviewers should follow certain guidelines
- Reviewers generally don't assess reproducibility
- Reviewers are generally not paid by the journal or conference (same goes for authors)
- Reviewers can relay confidential information to the editor not to be shared with the authors

Step 3: Peer-review

- Can be harsh (aka Reviewer 2 phenomenon)

“For a section on thought,
very little seems to have gone
into it.”

“But fundamentally, why did
you bother?”

“the authors seem to be
reinventing the wheel and a
flat tire to go along with it.”

Please don't do this when providing feedback

Step 3: Peer-review

Destructive feedback	Constructive feedback
The first sentence is unfortunate	You could make the first sentence more specific by [...]
The experimental design is a bit funny	The experimental design misses X and Y, you can add them to improve the paper
Line 306 .The sentence follows a bit of a Yoda-ish grammar	Line 306. I believe the sentence is not grammatically sound and can be rewritten as [...]

Step 3: Peer-review

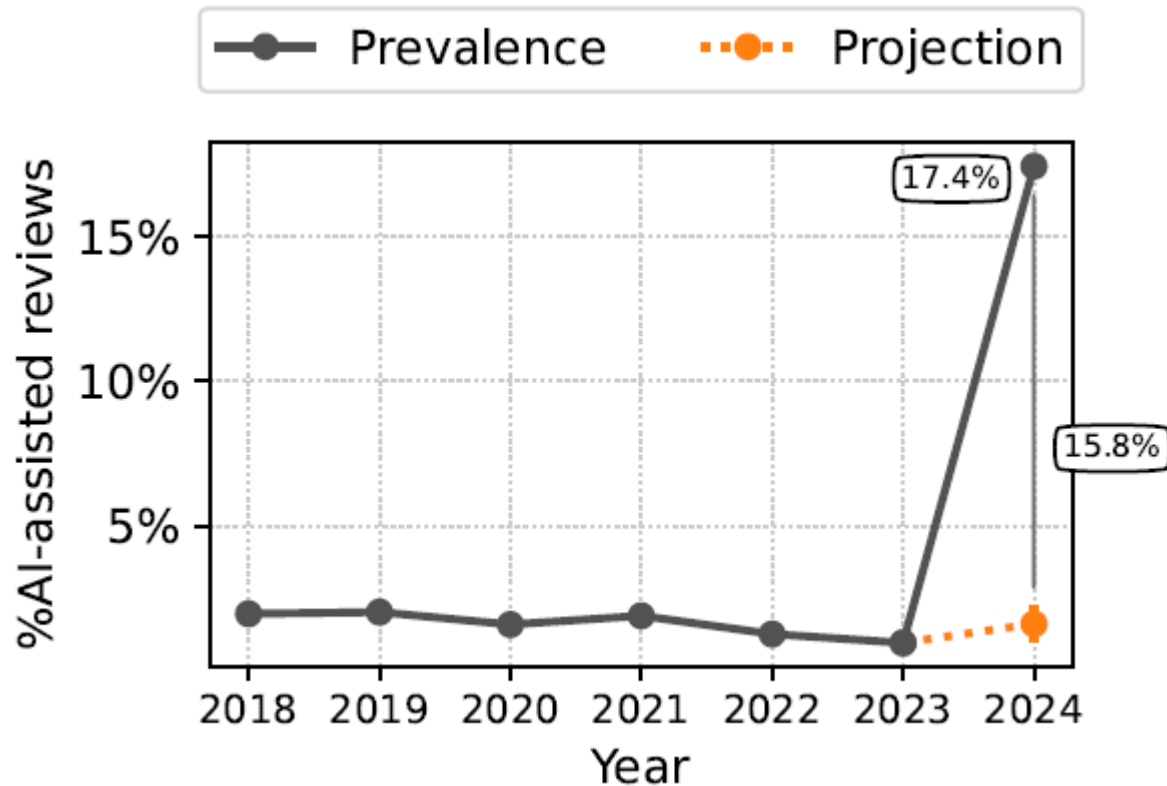
- Can also lead to “scooping” or other fraudulent practices

Computer scientist loses at least three papers, two for faked reviews

A journal waited 13 months to reject a submission. Days later, it published a plagiarized version by different authors

Step 3: Peer-review

■ Role of genAI





Step 3: Peer-review

Scientists hide messages in papers to game AI peer review

Some studies containing instructions in white text or small font – visible only to machines – will be withdrawn from preprint servers.

Contributions.

In short, we answer both questions fully. Specifically:

- (a) We prove an instance-specific high-probability lower bound on the clustering error rate for **MCC**. This reveals the problem-difficulty quantity $\mathcal{D}$: the minimum weighted KL divergence between the transition kernels (Section 3).
- (b) We propose a two-stage clustering algorithm that achieves near-optimal clustering error. Notably, it does not require any *a priori* knowledge of the underlying model, yet fully adapts to the given problem difficulty (Section 4). Especially for Stage I, we introduce a new injective Euclidean embedding specifically designed for ergodic Markov chains. This embedding, a contribution of independent interest, facilitates sharp concentration results for spectral clustering analysis (Section 4.1).
- (c) Our upper and lower bounds reveal gaps in misclassification errors and the required trajectory length H . Building on recent advances in concentration inequalities (Paulin, 2015; Fan et al., 2021) and estimation techniques (Wolfer and Kontorovich, 2021) for Markov chains, we elucidate the inherent complexities of clustering in **MMC** that currently render these gaps unavoidable (Appendix D).

Notation.

For a positive integer $n \geq 1$, let $[n] := \{1, 2, \dots, n\}$. For a set X , let $\Delta(X)$ be the set of probability distributions over X . Let $a \vee b := \max\{a, b\}$ and $a \wedge b := \min\{a, b\}$. We will utilize the asymptotic notations $\mathcal{O}, o, \Omega, \omega, \Theta$ freely throughout. For aesthetic purpose, we will also use $f \gtrsim g, f \lesssim g, f \asymp g$, defined as $f = \Omega(g), f = \mathcal{O}(g), f = \Theta(g)$, respectively.

2 Problem Setting



Step 4: Editorial decision

- Based on the peer-review reports, the editor makes a decision
 - ☐ Accept as is
 - ☐ Minor revision
 - ☐ Major revision
 - ☐ Reject

Step 4: Editorial decision

- Decisions by editors and reviewers can be driven by the outcome of the study
 - Significant effects or “positive results” (improvement on state of art) are more likely to be accepted
 - Leads to **publication bias** and the **file-drawer effect**



Step 5: Revision and response

- Authors revise their papers (depending on outcome Step 4)
 - Repeat Steps 2 – 4, until paper is accepted or rejected
 - Sometimes Step 3 is skipped when the revision is minor
- Often accompanied by response letter/rebuttal
 - Starts by thanking editor and reviewers
 - Specifies how changes are marked
 - Repeats *each* point raised by the reviewers followed by a response



Step 5: Revision and response

■ Example of response letter:

Dear Editor and Reviewers,

We would like to thank you for your careful reading of our manuscript and for the insightful and constructive comments. We have revised the manuscript accordingly and believe it has been significantly improved. Below, we provide a detailed, point-by-point response to each comment. All changes in the manuscript are highlighted in blue.

Reviewer #1

Comment 1:

The sample size appears small. Please justify its adequacy.

Response:

We appreciate this concern. We have added a justification for the sample size in the Methods section (p. 6). More specifically, sample size was based on...

Comment 2:

Instead of performing a paired samples t-test, I would consider an independent samples t-test.

Response:

Thank you for the suggestion. However, we believe an independent samples t-test would not be appropriate as the data come from the same subjects, thus creating a dependency.



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Quiz

- Which statement about peer review is correct?
 - ☐ Authors do not know the identity of reviewers
 - ☐ Peer review is always a constructive process
 - ☐ Reviewers usually rerun the analyses to check whether the results are correct
 - ☐ In some cases, reviewers plagiarize the manuscript they were tasked to evaluate



Quiz

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Quiz

- Researcher A estimates the accuracy of a model predicting Alzheimer's disease based on brain scans to be 82%. Researcher B tries to reproduce this result, but finds a lower performance (i.e., 71%). What could explain this?
 - ☐ Class imbalance
 - ☐ Fraud
 - ☐ Confound in training/validation/test split
 - ☐ All of the above

Quiz

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 - ☐ Confound in training/validation/test split
 - ☐ **All of the above**